

Has the Structure of Risk in Financial Markets Changed Since COVID-19?

REPORT N°2

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Abstract

We analyse whether the structure of risk in US financial markets has changed since the COVID-19 shock, using three key assets representing distinct risk categories – the S&P 500 equity index, gold and US investment-grade corporate bonds – on a daily observations sample from 2010 to 2026. We decompose and analyse risk structure in three layers: risk level (conditional volatility), risk co-movement (DCC correlations, empirical betas), and risk regime stability (parameter shifts, structural breaks). Univariate GARCH and EGARCH models are fitted by quasi-maximum likelihood with Bollerslev-Wooldridge robust standard errors paired with Wald tests for pre- vs. post-COVID parameter stability tests. We then estimate a Gaussian-QML DCC-(E)GARCH model and test for breaks in the resulting time-varying correlations. We find that the conditional volatility level spiked temporarily in 2020 but mean-reverted afterwards, while the cross-asset correlation structure shifted durably: the equity-bond correlation switched from -0.33 to +0.18 across COVID, and the realised raw Sharpe ratio of a representative 60/40 - S&P 500 / IG Bonds - portfolio collapsed from 1.04 to 0.54. Our answer to the research question is yes – the risk structure has changed since COVID-19. The shift is mostly concentrated in the cross-asset co-movements. For the equity index, volatility shocks became more persistent after COVID. For US IG Bonds, the variance process also shifted significantly, mainly through higher short-run shock sensitivity.

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1 Introduction

Beyond the sharp spike in volatility in March 2020, has COVID-19 left a lasting mark on the risk structure of financial markets? This question is critical for portfolio construction; indeed, the quality of a diversified portfolio depends heavily on how the selected assets move in tandem, and this is all the more important during times of crisis. Many assumptions have gradually taken root over three decades of negative correlation between stocks and bonds. While the environment that gave rise to the classic 60/40 portfolio has changed since 2020, the cost of ignoring this shift results in poor risk assessment and a sudden reduction in diversification. We address this issue using the dataset provided for the project and by focusing on three time series representing the equity, real asset, and fixed-income components of the U.S. financial markets: the S&P 500 Index, the gold denominated in USD and the U.S. investment-grade corporate bond index. We analyze the structure of risk across the following three dimensions:

1. The level of risk – the magnitude of risk, measured by conditional volatility.
2. The covariation of risk – the inter-asset dimension, measured by dynamic conditional correlations and the implied empirical betas of DCCs.
3. The risk regimes – the dimension of temporal stability, measured by parameter stability tests over the COVID period and Bai-Perron-type break detection on dynamic correlations.

Our preliminary answer is yes: risk has indeed undergone a structural shift in U.S. financial markets. Although the level of conditional volatility has returned to normal following the sharp rise linked to COVID-19, the persistence parameters of GARCH/EGARCH models for the pre- and post-COVID periods indicate a statistically significant regime shift in risk for both the S&P 500 and U.S. investment-grade bonds. The covariance dimension has also changed in a lasting way: the Pearson correlation between stocks and bonds has changed sign, shifting from -0.33 to +0.18. Dynamic conditional correlations (DCCs) have reached higher average levels, and Bai-Perron breakpoint tests identify persistent breakpoints in early March 2020. As for the Sharpe ratio of our 60/40 portfolio, it fell by nearly 50% following the COVID-19 crisis.

The 2020 global pandemic should be viewed more as a temporal marker, and not necessarily as the exact cause of the regime shift we are highlighting. The peak in volatility in March 2020 corresponds exactly to the type of one-off shock that GARCH dynamics are capable of absorbing without permanently increasing the long-term level of variance. What has changed structurally is the rate of mean reversion: post-COVID persistence, measured by the β parameter of the EGARCH model, rose from 0.950 to 0.983 for the S&P 500, meaning that volatility shocks subside much more slowly than before. For U.S. investment-grade bonds, the change is reflected instead in a stronger reaction to shocks (α) and a higher long-term volatility level. Beyond that, the most notable change appears to be the shift in the covariance structure, likely driven by inflation and rate surprises in 2021–2022. Our empirical analysis highlights this distinction in the data; the macroeconomic mechanism is consistent with it but is not formally tested in this article as the dataset does not contain series on inflation or policy rates.

2 Literature Review

The question of whether financial risk structure changes during crises is part of a long empirical tradition. Longin and Solnik [21] documented that correlations between international equity markets rise sharply during bear markets, a finding that challenged the static correlation assumptions embedded in mean-variance portfolio theory. Ang and Chen [3] formalised this observation, showing that correlations are systematically higher in down-markets than in up-markets, implying that diversification fails precisely when it is most needed.

2.1 The stock–bond correlation: hedge or co-mover?

The sign of the equity–bond correlation is the most influential input that underpins the traditional 60/40 portfolio construction. From the late 1990s until 2020 that correlation was negative, making bonds an effective hedge against equity drawdowns. 2022 then produced the worst calendar year for the 60/40 portfolio in modern history, with US Treasuries and the S&P 500 falling together. Campbell, Sunderam and Viceira [13] and Campbell, Pflueger and Viceira [12] state that there is a distinction to be made between demand shocks and supply shocks. Demand shocks dominate when global economic growth persists. In this case, stocks’ cash flows are strong and interest rates rise gradually as policymakers in a dynamic economy encourage tighter monetary policy. Hence, bond prices would fall while stocks move upwards, making the correlation negative. On the other hand, when supply-side economic shocks occur (e.g. geopolitical events or natural disasters) or when inflation exceeds the inflation target considerably, policymakers are pushed to adopt a more proactive/aggressive monetary policy, where they direct the economy. The large increase in interest rates to combat inflation impacts negatively both stock markets and bond prices, which is why we see positive correlation between both asset classes [24]. The literature therefore predicts a persistent structural break in the equity–bond correlation around the 2022 Fed tightening period, whereas the Covid-19 shock would correspond to a temporary break. McMillan [22] shows that the post-2022 sign reversal is not US-specific but is visible across six of the seven G7 economies.

2.2 Dating regime change: Bai–Perron, Hamilton, DCC

Hamilton [17, 19, 20] introduces regime-switching models in which an unobserved state follows a discrete Markov chain; the Hamilton filter delivers smoothed probabilities of being in each regime at every date. The key feature is the distinction between recurring and persistent regimes. A recurrent regime occurs when there is a low probability of remaining in a state between 2 time periods. On the contrary, a persistent regime occurs when there is a high probability of remaining in a state between 2 periods.

In practice, a two-state model can help explain different types of market turbulence. A high-volatility regime has a high transition probability back to the low-volatility regime is consistent with the Covid-19 crisis (2020). In contrast, it looks like the post-2022 period is closer to a permanent structural change in the market. Ang and Bekaert [2] and Ang and Timmermann [4] document that financial-market regimes are typically two-state with persistent low- and high-volatility regimes, coherent with recession dating of the NBER.

Engle [15] and Engle and Sheppard [16] develop the theoretical and empirical properties of the Dynamic Conditional Correlation (DCC) Multivariate GARCH model, a parsimonious framework for estimating large time-varying covariance matrices. The model decomposes the conditional covariance into univariate GARCH volatilities and a separately specified dynamic correlation matrix, enabling a two-stage estimation procedure in which the number of parameters requiring simultaneous estimation is $\mathcal{O}(1)$ regardless of portfolio size. After testing on up to 100 S&P 500 sector indices and DJIA stocks, they rejected constant correlation in every case and showed that DCC outperforms the industry-standard risk metrics on portfolio variance fit and residual diagnostics.

Nelson [23] introduces the Exponential GARCH (EGARCH) model to address three key limitations of standard GARCH specifications: their symmetric response to positive and negative shocks, the non-negativity constraints they impose on parameters, and the ambiguity of shock persistence. By modelling the logarithm of conditional variance as a linear function of past standardised innovations, the framework naturally enforces positive variance and disentangles sign (leverage) effects from magnitude effects.

In their research papers, Bai and Perron [5, 6] tackle a long-standing gap in the literature, as most prior work on structural breaks dealt with a single break at an unknown date. Instead, they proposed a dynamic-programming procedure that estimated and tested linear regression models with an unknown number of structural breaks at unknown dates, deriving the consistency, rate of convergence, and limiting distribution of the break-date estimators obtained by global least-squares minimization. The Bai–Perron framework has been widely adopted in empirical macroeconomic and financial research due to its ability to endogenously identify multiple structural changes in time series.

2.3 Gold as a Hedge and Safe Haven Asset

Baur and Lucey [7] define a hedge as an asset uncorrelated or negatively correlated with another on average, and a safe haven as one that displays such (non-)correlation specifically in stress periods. Using a GARCH regression of gold returns on stock and bond returns, they find that gold acts as a safe haven for stocks but not for bonds in the US, UK and Germany. Baur and McDermott [8] will later run a similar study but testing gold only on stocks, emphasizing on weak vs strong forms of safe havens, and will find that gold is a safe haven asset for the US and major European markets but not for emerging markets. Furthermore, Baur observed that this safe haven property was often visible in the short-term and that it could function as a tactical crisis asset rather than a long-term crash insurance instrument. A related stylised fact concerns the asymmetric response of volatility to return shocks. While equities exhibit the classical leverage effect of Black [10], where negative returns raise conditional volatility more than positive returns of the same magnitude, Baur [9] documents an inverse leverage effect in gold, with positive shocks generating larger volatility responses than negative ones, which is consistent with the safe-haven property. However, whether it retained these properties through the post 2020 period remains contested. Akhtaruzzaman et al. [1] and Cheema, Faff and Szulczyk [14] find that gold acted as a safe haven in the early pandemic but showed weaker or unstable performance thereafter.

3 Data

3.1 Sources and Coverage

The dataset used for our project is obtained from the provided Excel file (Data.xlsx) that contains observations of daily closing prices for various assets over the period ranging from 1990 to April 2026. In this project, we focus on the time series of the S&P 500 index, the Gold’s spot price and the US Investment Grade Bond index. We also restrain the sample period to observations from January 2010 to April 2026. We do this to ensure the pre-COVID observations reflect a stable post-Global Financial Crisis market state rather than a mix of several structural regimes.

3.2 Data Transformation

Before applying the models, we transform the selected time series into continuously compounded returns, expressed in percentage points:

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right) \times 100 \quad (1)$$

3.3 COVID-19 break point

We define the COVID-19 break point as February 19th, 2020, the day the S&P 500 reached its pre-crash peak. This choice is motivated by economic meaningfulness as it corresponds to the last date before markets seriously began pricing in the pandemic’s potential impacts on the economy.

3.4 Descriptive Statistics

Table 1: Descriptive statistics of daily log returns (%), post-GFC sample (2010–2026)

	S&P 500	Gold	US IG Bonds
Observations	4101	4236	4267
Mean	0.0450	0.0344	0.0152
Std.Dev.	1.0909	1.0122	0.3200
Median	0.0698	0.0487	0.0272
Skewness	-0.6069	-0.6452	-0.7894
Excess Kurtosis	13.4736	6.4795	8.6564

The S&P 500 exhibits the highest daily volatility (1.091%) and leptokurtosis, while Gold’s volatility is marginally lower (1.012%). Equities’ mean return (0.045%) aligns with the post-Global Financial Crash bull market. US IG Bonds, as expected for fixed income, show significantly lower volatility (0.320%) and returns (0.015%). All three series display left-tailedness, which is typical for financial returns.

3.5 Aligned Sample

The three series are all denominated in USD, making their returns directly comparable. They do not, however, share identical trading calendars. The descriptive statistics and the univariate GARCH/EGARCH estimations use each asset’s own available sample. For the DCC step only, we align the three price series on common trading dates before computing log returns. This ensures that every return is computed over the same holding period. The aligned sample yields 4,101 common observations. The S&P 500 has the fewest trading days of the three series, and loses no observations. Gold and Bonds lose 135 and 166 observations respectively, producing insignificant differences in their descriptive statistics relative to the individual series.

4 Methodology

4.1 Pre-Estimation Diagnostics

Before building the models, we apply two standard diagnostics. The Augmented Dickey-Fuller (ADF) test to confirm stationarity of the return series (H_0 : unit root) and the Ljung-Box test on squared returns to test for ARCH effects in the data (H_0 : no autocorrelation in squared returns). Rejection of the null-hypotheses for both tests suggest GARCH-type modelling is appropriate.

4.2 GARCH(1,1) Model

The symmetric GARCH model proposed by Bollerslev [11] allows to capture volatility clustering by estimating variance using both past squared shocks and previous variance. By decomposing returns into a constant mean and a conditionally heteroskedastic error, one gets that the best approximation for variance is squared errors. Building an AR(1) on that gives the well-known ARCH(1). Adding the first lag variance gives the full GARCH(1,1) equation:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (2)$$

where $\omega > 0$ is the constant term, $\alpha \geq 0$ measures the reactivity of volatility to recent shocks, $\beta \geq 0$ measures the persistence of past conditional variance, and $\alpha + \beta < 1$ for weak stationarity, captures total persistence. The innovations z_t follow a standardised Student-t distribution with

estimated degrees of freedom ν , jointly estimated alongside the volatility parameters. The long-run annualised volatility, assuming 252 trading days, is calculated as:

$$\bar{\sigma} = \sqrt{\frac{\omega}{1 - \alpha - \beta} \times 252} \quad (3)$$

4.3 EGARCH(1,1) Model

The EGARCH model of Nelson [23] allows to capture the asymmetry in volatility responses between positive and negative shocks on individual returns by modelling the log of conditional variance:

$$\ln(\sigma_t^2) = \omega + \alpha(|z_{t-1}| - \mathbb{E}|z|) + \gamma z_{t-1} + \beta \ln(\sigma_{t-1}^2) \quad (4)$$

where $z_{t-1} = \varepsilon_{t-1}/\sigma_{t-1}$ is the standardised innovation, assumed to follow a standardised Student-t distribution. Because the model is written for $\ln \sigma_t^2$, positivity of conditional variance holds by construction and no non-negativity constraints on (ω, α, β) are required. The parameter α captures the magnitude effect and γ captures the sign asymmetry: $\gamma < 0$ indicates the leverage effect, negative shocks increase volatility more than positive shocks of equal size.

4.4 Model Selection: GARCH vs. EGARCH

Since GARCH and EGARCH model variance differently, the first models it directly while the latter models its logarithm, we compare them using the Akaike and Bayesian Information Criteria within the same sample period:

$$AIC = -2\ell + 2k, \quad BIC = -2\ell + k \ln(T) \quad (5)$$

where ℓ is the maximised log-likelihood, k the number of estimated parameters, and T the number of observations; BIC penalises complexity more strictly than AIC. We refine the choice with two complementary criteria: the statistical significance of the EGARCH leverage coefficient γ (a non-significant γ undermines the case for the asymmetric specification), and the Ljung–Box test on squared standardised residuals $\hat{z}_t^2$ at lags 5, 10 and 20, which directly tests whether the variance equation has absorbed the conditional heteroskedasticity.

4.5 Robust Wald Test, Distribution Adequacy, and Residual Diagnostics

Because we treat the Student- t likelihood as a working approximation, we estimate the variance parameters via quasi-maximum likelihood. Under QML, Bollerslev and Wooldridge [18] show that the sandwich covariance $\hat{V}_{BW} = A^{-1}BA^{-1}$ produces correct inference even when the conditional density is misspecified, thus supporting our choice. To test whether the parameters shifted at the chosen COVID break (19 February 2020), we apply a Chow-style Wald test, $H_0 : \theta_{\text{pre}} = \theta_{\text{post}}$, built on these robust covariances:

$$W = (\hat{\theta}_{\text{pre}} - \hat{\theta}_{\text{post}})^\top (\hat{V}_{\text{pre}} + \hat{V}_{\text{post}})^{-1} (\hat{\theta}_{\text{pre}} - \hat{\theta}_{\text{post}}) \sim \chi^2(k), \quad (6)$$

which is asymptotically χ^2 even under distributional misspecification. We test the variance vector $\theta = (\omega, \alpha, \beta)$ for GARCH(1,1) and $\theta = (\omega, \alpha, \gamma, \beta)$ for EGARCH(1,1); ν is excluded from the joint test because it parametrises the working density rather than the variance equation. The adequation of the distribution is checked by a Kolmogorov–Smirnov test of the rescaled standardised residuals against the fitted Student- $t(\hat{\nu})$, and residual ARCH by Ljung–Box on $\hat{z}_t^2$ at lags 5, 10 and 20.

4.6 DCC-(E)GARCH Model — Mixed Specification

The Dynamic Conditional Correlation (DCC) model of Engle and Sheppard [16] allows to capture time-varying correlations between the three selected variables. We implement it with a mixed specification: EGARCH(1,1) for the S&P 500 and GARCH(1,1) for Gold and US IG Bonds. AIC and BIC alone prove this choice for the S&P 500 (EGARCH wins) and US IG Bonds (GARCH wins). For Gold, the two criteria slightly favour EGARCH, but the Ljung–Box test on $\hat{z}_t^2$ rejects EGARCH much more strongly than GARCH, therefore GARCH is chosen. The DCC procedure can be summarised into the following steps.

Step 1 — Volatility Models

We estimate the appropriate univariate model for each asset on the aligned common sample, yielding standardised residuals:

$$\hat{u}_{i,t} = \hat{\varepsilon}_{i,t} / \hat{\sigma}_{i,t}, \quad \text{where } i \in \{\text{S\&P500, Gold, Bonds}\}. \quad (7)$$

The aligned-sample diagnostics are not perfect: the S&P 500 residuals pass the Ljung–Box tests on squared standardised residuals, while Gold and US IG Bonds retain short-lag residual ARCH effects. We therefore treat the DCC estimates as informative time-varying correlation measures, but interpret especially the Gold-related correlations with caution.

Step 2 — DCC Estimation

The conditional covariance matrix is decomposed as $\Sigma_t = D_t^{1/2} \Gamma_t D_t^{1/2}$, where Γ_t , the time-varying correlation matrix, is obtained by standardising the auxiliary matrix Q_t , which follows the recursion $Q_t = (1 - \alpha - \beta)\bar{Q} + \alpha u_{t-1} u'_{t-1} + \beta Q_{t-1}$, where $\bar{Q} = \mathbb{E}[u_t u'_t]$. The two scalar parameters (α, β) are estimated by maximising the DCC log-likelihood via SLSQP, subject to $\alpha, \beta > 0$ and $\alpha + \beta < 1$ with starting values $\alpha_0 = 0.05$ and $\beta_0 = 0.90$. We report point estimates and focus on the economic interpretation of the resulting dynamic correlations.

4.7 DCC-Implied Empirical Betas

The DCC-implied beta of asset i relative to the S&P 500 is $\beta_{i,t} = \rho_{i,SP500,t} \cdot \sigma_{i,t} / \sigma_{SP500,t}$. It measures how much each asset’s return co-moves with the S&P 500 at each point in time, capturing whether the asset effectively acts as a hedge. We compute it for Gold and US IG Bonds using the full-sample DCC correlations and aligned conditional volatilities, yielding a continuous daily series from 2010 to 2026.

4.8 Structural Break Analysis — Bai-Perron

We apply the Bai-Perron [5] test to each of the three DCC correlation series to detect shifts in mean correlation levels, fitting a pure mean-shift model $y_t = \mu_j + \varepsilon_t$ within each segment. Break dates are estimated via dynamic programming with an LWZ penalty $(k(\ln T)^2)$, allowing up to 3 breaks with a minimum segment length of 126 trading days.

4.9 60/40 Portfolio Analysis

We construct a daily-rebalanced 60/40 portfolio — $r_t^{60/40} = 0.60 \cdot r_t^{SP500} + 0.40 \cdot r_t^{Bonds}$ — whose diversification logic rests on the historically negative equity-bond correlation. We evaluate it against its individual components on annualised return, volatility, Sharpe ratio, and maximum drawdown across the pre- and post-COVID subsamples, and track the diversification ratio to quantify how much risk-reduction benefit was lost as the correlation turned positive.

5 Empirical results

5.1 Univariate Volatility Analysis

Initially, we focus on each asset class separately. For each asset, we test which model is best suited to estimate its conditional volatility. To do this, we compare two models: the standard GARCH model and the exponential EGARCH model. By doing so, we seek to obtain a significant $\hat{\gamma}$ coefficient that would be characteristic of an asymmetric response to a shock of the same intensity, depending on the sign of the shock. This phenomenon is more commonly referred to as the leverage effect.

Table 2: GARCH(1,1) vs. EGARCH (1,1) Parameters Comparison - S&P 500

Sample	Model	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\gamma}$ (Asym)	$\hat{\beta}$	LR Vol (%)
Full Sample (2010 - 2026)	GARCH	0.0252	0.1635	-	0.8266	25.30
	EGARCH	-0.0091	0.1715	-0.1967	0.9660	13.88
Pre-COVID (2010 - Feb 2020)	GARCH	0.0263	0.1772	-	0.8098	22.53
	EGARCH	-0.0239	0.1711	-0.2451	0.9500	12.50
Post-COVID (Feb 2020 - 2026)	GARCH	0.0320	0.1248	-	0.8523	18.76
	EGARCH	-0.0083	0.0590	-0.1647	0.9827	12.49

An analysis of our initial results for the stock market reveals a significant and negative $\hat{\gamma}$ coefficient regardless of the observation period. This result confirms the leverage effect highlighted by Nelson [23], indicating that negative innovations lead to significantly higher volatility than positive shocks of comparable magnitude. We note that the information criteria (AIC, BIC) are consistently lower for the EGARCH model, along with clear significance of the gamma parameter and appropriate residuals, thus agreeing with our pre-selection of EGARCH for the S&P500.

When comparing the pre- and post-pandemic subsamples, the estimated parameters reinforce our view that there has been a structural change in market risk. We immediately observe a dampening of the asymmetric response, with $\hat{\gamma}$ moving from -0.2451 before COVID to -0.1647 after COVID, falling in magnitude. At the same time, the coefficient ($\hat{\alpha}$), which measures responsiveness to short-term shocks, decreases from 0.1711 to 0.059. This suggests that the market has become less sensitive to sudden volatility spikes driven by news events. On the other hand, the persistence coefficient ($\hat{\beta}$) has risen from 0.9500 to 0.9827 in the recent sample. This suggests an increase in persistence, or a greater memory of volatility. The market is less sensitive to stress, but once doubt sets in, it lasts longer.

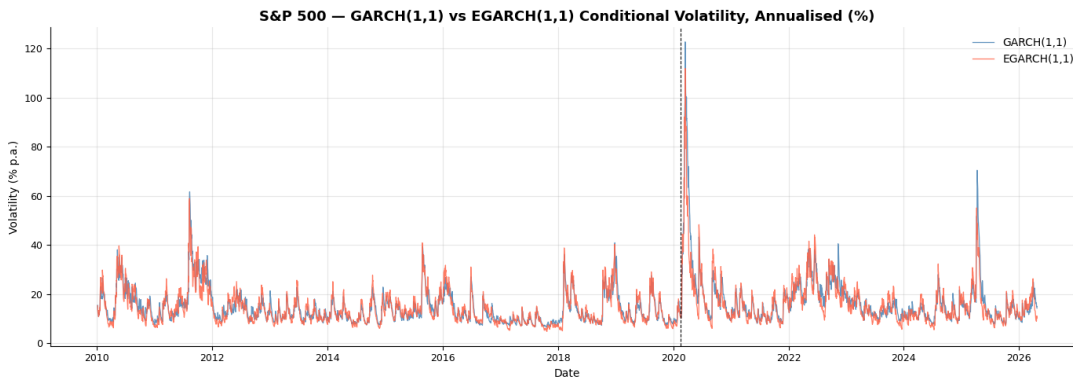


Figure 1: S&P 500 — GARCH(1,1) vs EGARCH(1,1) Conditional Volatility, Annualised (%)

Table 3: GARCH(1,1) vs. EGARCH (1,1) Parameters Comparison - GOLD

Sample	Model	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\gamma}$ (Asym)	$\hat{\beta}$	LR Vol (%)
Full Sample (2010 - 2026)	GARCH	0.0150	0.0499	-	0.9368	16.91
	EGARCH	0.0073	0.1194	0.0328	0.9847	20.18
Pre-COVID (2010 - Feb 2020)	GARCH	0.0076	0.0342	-	0.9575	15.23
	EGARCH	0.0045	0.0959	0.0170	0.9903	19.98
Post-COVID (Feb 2020 - 2026)	GARCH	0.0484	0.0802	-	0.8785	17.19
	EGARCH	0.0107	0.1479	0.0508	0.9729	19.34

Applying the same univariate analytical framework to gold reveals a risk structure that differs from that of the equity markets. The EGARCH coefficients indicate a slightly positive asymmetry parameter: 0.0170 Pre-COVID and 0.0508 Post-COVID. Thus, unlike the S&P 500, gold exhibits a slight increase in volatility following an upward price movement. This behaviour is largely consistent with the common perception of gold as a safe-haven asset that holds up well during periods of uncertainty. However, EGARCH is not our working specification for Gold in the DCC stage because its squared standardised residuals retain stronger short-run ARCH effects than under GARCH. In the symmetric GARCH model, the short-run shock coefficient ($\hat{\alpha}$) rises from 0.0342 to 0.0802, while volatility persistence ($\hat{\beta}$) falls from 0.9575 to 0.8785. From an economic perspective, this implies that the post-crisis regime, marked by higher inflationary pressures and a restrictive monetary policy, has made the gold market less stable and therefore more sensitive to macroeconomic news. Nevertheless, the robust Wald test below does not support a statistically significant break in Gold's variance parameters.

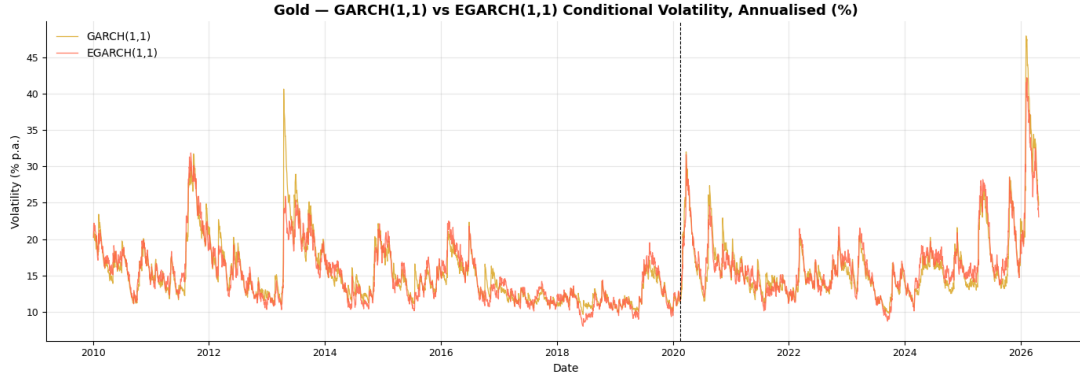


Figure 2: Gold — GARCH(1,1) vs EGARCH(1,1) Conditional Volatility, Annualised (%)

Table 4: GARCH(1,1) vs. EGARCH (1,1) Parameters Comparison - US IG Bonds

Sample	Model	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\gamma}$ (Asym)	$\hat{\beta}$	LR Vol (%)
Full Sample (2010 - 2026)	GARCH	0.0010	0.0519	-	0.9375	4.89
	EGARCH	-0.0207	0.0950	-0.0221	0.9905	5.32
Pre-COVID (2010 - Feb 2020)	GARCH	0.0005	0.0283	-	0.9639	4.04
	EGARCH	-0.0197	0.0661	-0.0003	0.9922	4.53
Post-COVID (Feb 2020 - 2026)	GARCH	0.0023	0.0705	-	0.9131	5.91
	EGARCH	-0.0225	0.0736	-0.0504	0.9892	5.59

Finally, for our third and final asset class—investment-grade bonds issued by U.S. corporations—the individual risk profile is yet another story. Because the selected working model for US IG Bonds is GARCH, the main interpretation come from $\hat{\omega}$, $\hat{\alpha}$ and $\hat{\beta}$. Prior to COVID, these bonds were characterised by low responsiveness to innovations, with $\hat{\alpha} = 0.0283$, and high persistence, with $\hat{\beta} = 0.9639$. After COVID, the sensitivity to short-term shocks rises to $\hat{\alpha} = 0.0705$, while persistence falls to $\hat{\beta} = 0.9131$. This means the bond result is not mainly a story of higher persistence. It is a story of stronger immediate reaction to shocks and a higher long-term annualised volatility, rising from 4.04% to 5.91%. The EGARCH estimates also show a moderate negative asymmetry post-COVID that may have some consequences on its role as a diversifier, but this is secondary because GARCH is the retained specification. We analyse this impact on the performance of a hypothetical 60/40 portfolio in Section 5.4.

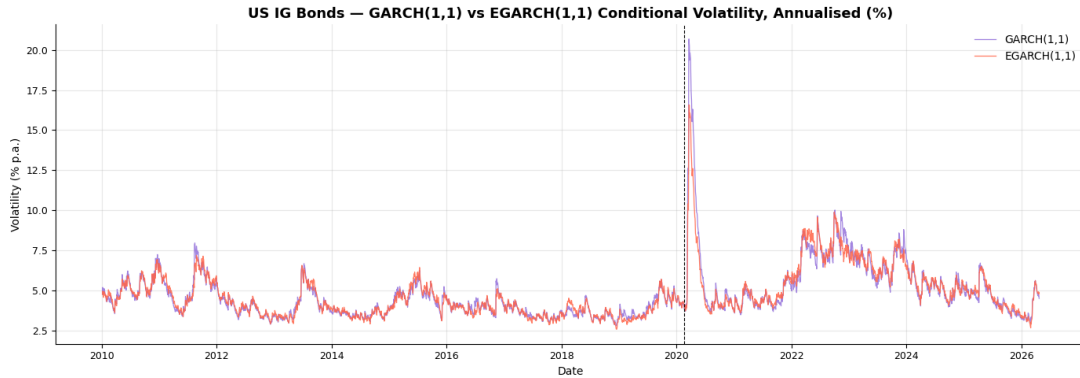


Figure 3: US IG Bonds — GARCH vs EGARCH Conditional Volatility, Annualised (%)

In order to confirm our interpretations of the changes observed between the two periods as defined above, we conduct a robust Wald parameter stability test (see Table 5).

First, for the equity market modeled by EGARCH to account for leverage, we observe that the null hypothesis of parameter constancy is rejected (p-value = 0.0018). This result proves that there is indeed a before-and-after distinction in the equity risk regime. The same holds true for U.S. IG bonds (p-value = 0.0117), confirming the structural break mentioned earlier. As for commodities, and more specifically gold, it is interesting to note that a Wald test fails to reject the stability of the parameters (p-value = 0.1708), even at a low significance level (i.e., a 10% threshold). All despite the fact that the descriptive parameters suggest increased volatility; the robust statistical test confirms that gold’s risk structure has remained similar despite the crisis.

Table 5: Robust Wald parameter stability Tests

Asset	Model	Wald Stat	p-value	Decision (H_0 : Stability)
S&P 500	EGARCH(1,1)	17.174	0.0018 ***	Reject H_0 (Break)
US IG Bonds	GARCH(1,1)	11.011	0.0117 **	Reject H_0 (Break)
Gold	GARCH(1,1)	5.014	0.1708	Fail to reject H_0 (Stable)

Before moving to the DCC step, the residual diagnostics need one qualification. The S&P 500 EGARCH residuals are well behaved, but Gold and US IG Bonds retain some residual ARCH effects in the full-sample diagnostics. For Gold, the issue is mainly concentrated in the full and pre-COVID samples, while the post-COVID GARCH residuals no longer reject the Ljung–Box null. The ACF of Gold’s squared standardised residuals, reported in Figure 9, also shows that the remaining dependence is mostly a very short-run issue, concentrated around the first lag rather than a persistent pattern across many lags. For US IG bonds, the full-sample GARCH residuals reject the null, but the pre- and post-COVID samples separately do not. This pattern gives

useful context: for bonds, the failure of the full-sample diagnostic is consistent with a parameter break, since a single GARCH process is forced to average across two regimes. For Gold, it suggests that the simple GARCH(1,1) still leaves some short-run volatility dependence, so the Gold-related DCC correlations should be interpreted more cautiously. The main equity–bond result is however less affected because it is also supported by the realised Pearson correlation, the DCC averages, the Bai–Perron break and the 60/40 portfolio deterioration.

5.2 DCC: Cross-Asset Dynamics

Now that our models for estimating the univariate volatility of each series have been specified, we can apply the two-stage DCC-(E)GARCH framework developed by Engle [15]. We take care of course to select a time-aligned sample with 4,101 daily observations for our three series to avoid any temporal approximation in the multivariate system. The first empirical estimation of this model yields correlation parameters $\hat{\alpha}_{DCC} = 0.0309$ and $\hat{\beta}_{DCC} = 0.9564$, resulting in a total persistence of 0.9873 for our model. Since not all models are of the EGARCH type, the coefficient $\hat{\gamma}_{DCC}$ is naturally excluded. These results confirm that the relationships between our three assets are indeed dynamic in the first instance, but also that they are characterised by substantial memory. Upon a more rigorous analysis of these dynamic correlations presented in Table 6 below, we find this profound structural break that was already present in the univariate volatility of our assets. This break is all the more pronounced within the relationship between stocks and bonds.

Table 6: Average dynamic conditional correlation ($\hat{\rho}$)

Asset Pair	$\hat{\rho}_{FullSample}$	$\hat{\rho}_{PreCOVID}$	$\hat{\rho}_{PostCOVID}$	$\Delta\hat{\rho}$
S&P 500 / US IG Bonds	-0.1261	-0.2460	+0.0706	+0.3166
S&P 500 / Gold	0.0487	0.0173	+0.1002	+0.0829
Gold / US IG Bonds	0.2806	0.2728	0.2934	+0.0206

The most interesting finding in our analysis of dynamic correlations is, as mentioned earlier, this break in the relationship between the stock and bond markets. Indeed, when we compare the average correlation between the pre-pandemic period (2010–2020) and the post-crisis period, we see a shift from a $\hat{\rho}$ of -0.2460 to +0.0706, representing a $\Delta\hat{\rho}$ of +0.3166. From an economic perspective, the implications of this shift are significant: bonds, often viewed as a solid stabiliser in mixed portfolios, now move in tandem with the stock markets (S&P 500). This undermines their role as a hedge, especially during market downturns. That transformation fits within the theoretical framework of Campbell, Sunderam, and Viceira [13], which points to a transition towards a supply-shock regime characterised by higher inflation and aggressive monetary tightening by the Fed. This context forces to reclassify US investment-grade bonds from their status as a diversifying hedge to a procyclical asset that moves in step with the market, albeit with less intensity.

Our second observation is that changes in the risk structure are also reflected in gold’s performance, especially through its co-movement with the equity index. Indeed, when we examine its relationship with the S&P 500 index prior to the COVID-19 crisis, we observe an average correlation $\hat{\rho}$ close to zero, specifically +0.0173, which made gold a strategically uncorrelated diversification tool. Yet, in the period that followed, its dynamic correlation $\hat{\rho}$ with stock markets increased to +0.1002. This increased co-movement with stocks questions gold’s traditional status as a safe-haven asset, which is consistent with recent literature (e.g., Cheema et al. [14]), particularly when the economy is in an inflationary environment or when the U.S. dollar is more volatile.

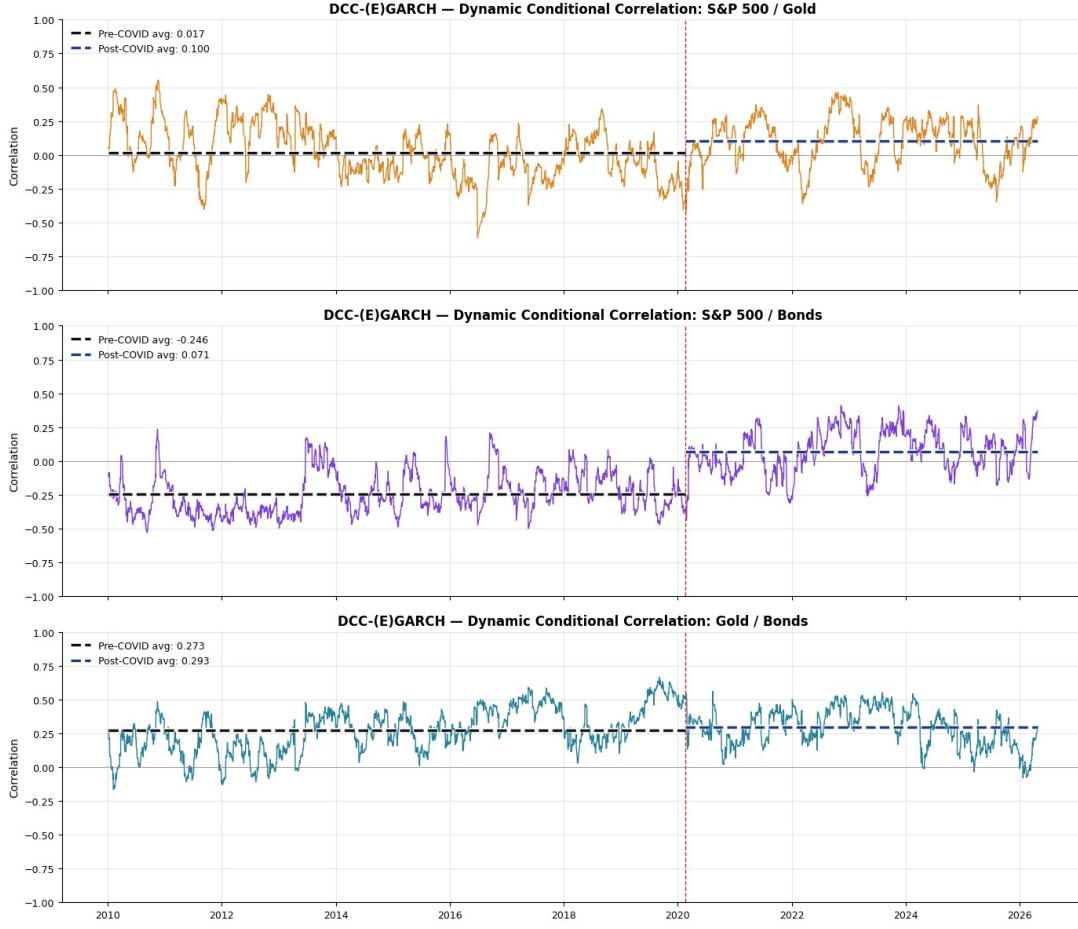


Figure 4: DCC-(E)GARCH — Dynamic Conditional Correlation for the three pairs

However, this is not a formal rejection of Gold’s safe-haven property, because a safe haven is defined by behaviour during stress periods rather than by average correlation over a whole sub-sample. Nonetheless, Figure 4 shows that around the COVID break and during several post-COVID episodes, the DCC correlation rises sharply, indicating that gold increasingly co-moved with equities precisely when macro-financial uncertainty was elevated.

Time-Varying Betas and Systemic Risk Exposure

In order to combine our two sections on univariate volatility $\hat{\sigma}_{i,t}$ (specific to the asset under study) and its dynamic correlation $\hat{\rho}_{i,m,t}$ with other assets, we estimate the relevant dynamic betas $\hat{\beta}_{i,m,t}$ using the S&P 500 index as the market benchmark. By doing so, we can examine the variation in the average beta across periods and isolate the exact factor driving the structural change.

Table 7: DCC Implied Betas relative to the S&P 500 (i.e. $\hat{\beta}_{i,S\&P500,t}$)

Asset	$\hat{\beta}_{PreCOVID}$	$\hat{\beta}_{PostCOVID}$	$\Delta\hat{\beta}$	$\Delta\hat{\rho}$	$\Delta(\hat{\sigma}_i/\hat{\sigma}_m)$
Gold	0.0327	0.1128	0.0801	0.0829	-0.0605
US IG Bonds	-0.0824	0.0340	0.1164	0.3166	0.0445

These results reinforce our previous interpretations regarding the collapse of the traditional defensive framework. Indeed, what stands out first are the average beta differentials $\Delta\hat{\beta}_{i,S\&P500,t}$

between the two periods, which is positive for both of our assets. If we focus first on US IG bonds, we observe a change of +0.1164, which is mainly due to the change in average correlation $\Delta\hat{\rho} = +0.3166$ rather than to a surge in the intrinsic volatility of bonds relative to stocks (this ratio having increased only marginally, by +0.0445). This reinforces our belief that the risk profile of bonds has changed due to a structural macroeconomic alignment with equities (i.e. supply shocks) rather than just increased volatility in bond markets. Now, if we look at the situation for gold, we can identify a similar trend, namely an increase in beta: $\Delta\hat{\beta}_{i,S\&P500,t} = +0.0801$. The beta decomposition adds a different layer to the previous correlation analysis. This increase does not come from higher relative volatility, since the volatility ratio actually decreases by -0.0605. Instead, it is almost entirely explained by the rise in Gold’s average correlation with the S&P 500.

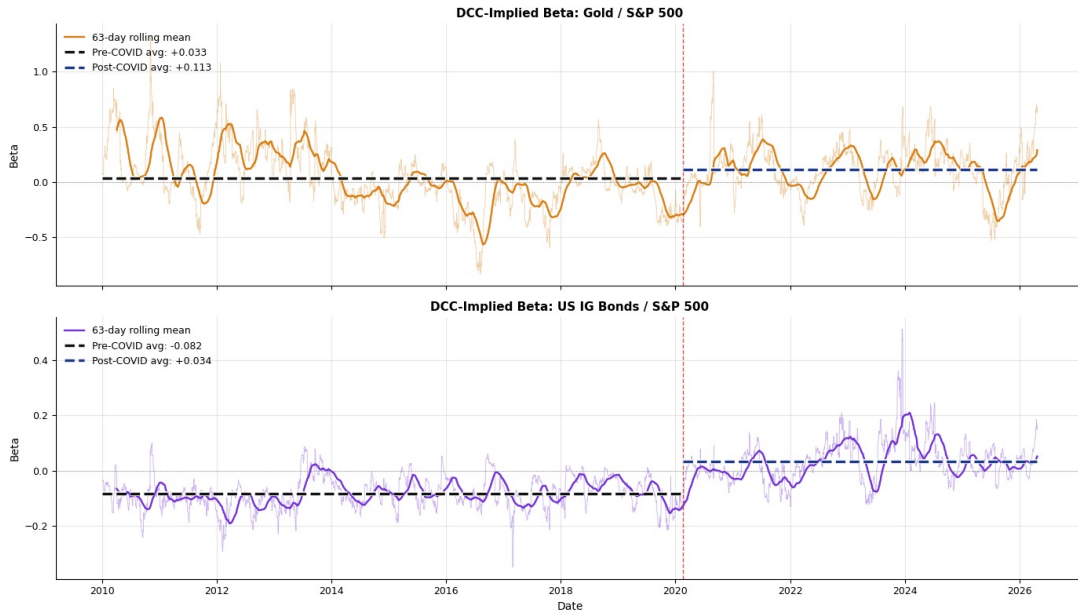


Figure 5: DCC-Implied Beta for Gold and US IG Bonds relative to the S&P 500

5.3 Endogenous Break Detection : Bai-Perron Test

While the peak in February 2020 provides a solid anchor point for dividing our subsample, imposing an exogenous break date is not very rigorous. To validate our hypothesis of a correlation reversal endogenously, we apply the framework developed by Bai and Perron [5, 6].

Table 8: Bai-Perron Test for endogenous break detection with change in mean $\hat{\rho}$.

Asset Pair	Break Date	$\hat{\rho}_{Pre-Break}$	$\hat{\rho}_{Post-Break}$
S&P 500 / US IG Bonds	March 10 th , 2020	-0.2193	+0.0740
S&P 500 / Gold	March 25 th , 2020	-0.0738	+0.1269
Gold / US IG Bonds	March 2 nd , 2020	+0.5296	+0.3433

Note: Pre-break and post-break means refer to the specific segments immediately surrounding the March 2020 structural break.

The empirical results provide striking statistical validation of the pandemic’s impact on the relationship between stocks and bonds. Our Bai-Perron test identifies several correlation regimes, one of which occurs around March 2020 across all three asset pairs. This supports the interpretation that COVID coincided with an important correlation-regime shift, it is also important

to report that the DCC correlations display additional breaks before and after COVID. For the S&P 500 / US IG Bonds pair, the average correlation coefficient $\hat{\rho}$ for the segment preceding the break was negative (-0.2193). Following the break, it shifted into positive territory (+0.0740). This is economically important and consistent with the weakening of the traditional diversification strategy. Our Bai-Perron test also highlights structural breaks in gold’s multivariate relationships during March 2020, specifically on March 25th for the S&P/Gold pair and on March 2nd for Gold/Bonds. The correct interpretation is not that COVID mechanically caused one permanent break, but that March 2020 is the most economically relevant common regime marker in a broader sequence of correlation shifts. In Section 6.2 on limitations, we provide further details on the Bai-Perron test, which is purely backwards-looking. It is possible that, at the very long-term macroeconomic level, what our model currently describes as a “permanent structural break” may, in reality, be nothing more than a “persistent regime shift,” echoing Hamilton [17, 19].

5.4 Portfolio Implications: The 60/40 Benchmark

To quantify the true economic magnitude of this correlation regime shift, we evaluate the performance of a traditional 60/40 portfolio (60% S&P 500, 40% US IG Bonds, daily rebalanced) across the pre- and post-COVID sub-samples. The 60/40 allocation is the foundational benchmark of institutional asset management, relying heavily on the historically negative equity-bond correlation to minimise portfolio variance and cushion drawdowns.

Table 9: Portfolio metrics before and after Break Date

Metric	S&P 500 (Pre)	S&P 500 (Post)	IG Bonds (Pre)	IG Bonds (Post)	60/40 (Pre)	60/40 (Post)
Ann. Return (%)	10.22	10.48	5.70	1.08	8.75	7.18
Ann. Volatility (%)	14.75	20.86	4.18	6.49	8.45	13.22
Sharpe Ratio	0.69	0.50	1.36	0.17	1.04	0.54
Max Drawdown (%)	-20.70	-35.86	-6.51	-21.93	-12.48	-26.86
Realized Correlation	–	–	–	–	-0.330	+0.176
Diversification Ratio	–	–	–	–	1.246	1.143

The results demonstrate a real breakdown in the portfolio’s risk-efficiency. The headline finding is the collapse of the 60/40 Sharpe ratio from 1.04 to 0.54 (-48%). Crucially, this deterioration was not driven by poor equity performance, the S&P 500 actually generated slightly higher annualised returns in the post-COVID regime (10.48% vs 10.22%). The Sharpe Ratio collapse therefore reflects a true diversification failure. As the realised Pearson correlation flipped from a defensive -0.330 to a pro-cyclical +0.176, the portfolio’s diversification ratio fell from 1.246 to 1.143. Consequently, the fixed-income allocation failed to act as a reliable stabiliser during market shocks. This exposed the portfolio to significantly deeper tail risks, with the Maximum Drawdown more than doubling from -12.48% pre-COVID to -26.86% post-COVID. Ultimately, this confirms that the structural regime shift identified in our econometric models (DCC and Bai-Perron) has profound, tangible implications for asset allocation: the automatic hedge of the 60/40 portfolio is materially weaker in the post-COVID sample.

6 Discussion

6.1 Economic intuition and portfolio implications

The transition from a macroeconomic environment dominated by demand shocks (characteristic of the 2010s) to a post-COVID regime of supply shocks has fundamentally reshaped the dynamics between these factors. In a new inflationary regime, restrictive monetary policies and discount rate adjustments have become the primary sources of fluctuations. This new economic context notably explains the observed shift in the equity-bond correlation, moving from a negative to a positive regime. But a prominent macroeconomic risk also helps explain the reduction in leverage effect captured by our EGARCH model. Unlike idiosyncratic shocks, which could stem from poor quarterly results and trigger asymmetric panic movements. Inflation shocks, on the other hand, penalise the valuation of the entire market via the risk-free rate, thereby generating a more symmetrical volatility reaction. Consequently, in the face of declines in both equity and bond markets, the traditional 60/40 allocation is structurally compromised and loses its defensive effectiveness. Our DCC models clearly demonstrate that navigating an inflationary environment requires a shift toward assets capable of maintaining an independent relationship with this systematic risk. In this regard, gold has proven its resilience and its role as a true portfolio stabilizer in times of uncertainty.

6.2 Limitations of the models used

The univariate GARCH family is too narrow

The GARCH(1,1) and EGARCH(1,1) specifications assume that the conditional variance depends only on the most recent squared innovation and variance. This captures volatility clustering at high frequency but fails to do so on longer-term. FIGARCH and GARCH-MIDAS specifications, allow for short- and long-term variance splitting as well as identifying longer-term volatility regimes of high frequency time series, by integrating macro variables. This could make the results more economically interpretable for the dynamics of volatility clusters.

EGARCH is very sensitive to small differences in asymmetry. The GJR-GARCH specification captures it more linearly by adding a dummy binary variable that introduces a linear threshold for the leverage effect, enabling the model to behave better when the leverage parameter is small or imprecise. Changing this would enable us to identify whether the positive leverage effect in post-COVID Gold is due to the nature of Gold as a defensive asset or the result of EGARCH's log-variance structure when asymmetry is low.

The Student- t innovation distribution captures fat tails but stays symmetric, while the skewness of each return series is negative. Our diagnostic tests also display some residual ARCH dependencies, especially for Gold and US IG bonds in the aligned sample, indicating that the models did not extract all short-run volatility dynamics. A generalised error distribution or a skewed- t distribution could allow to adjust the shape and skewness of the innovation distribution.

The DCC framework imposes too much homogeneity

The DCC-(E)GARCH model assumes that all asset pairs react in the same way to a shock and with the same persistence. This tends to smooth out sudden shocks and overstate slower shifts, resulting in biased correlation series. The Bai-Perron test contradicts this assumption, but cannot fully quantify the extent of the bias. Asymmetric DCC, flexible DCC or regime-switching DCC models would all allow pair-specific dynamics.

Misspecifications in the univariate GARCH/EGARCH step propagate into residuals and bias DCC parameters and correlations. Particularly with Gold, due to leverage and residual ARCH effects. This results in a cleaner correlation shift than what it statistically is. Re-estimating the full two-step process using a block-bootstrap procedure would allow to preserve temporal

dependency and construct confidence intervals around the correlation series. This would enable to distinguish structural breaks from estimation noise.

Additionally, the DCC model also assumes mean-reversion for conditional correlations. This contradicts empirical evidence of persistent, asymmetric correlation surges during crises. Therefore, the model is mechanically unable to capture the shift to a new long-term equilibrium, leaving temporary deviations as descriptive rather than economically interpretable findings.

The break-test machinery has its own assumptions

The Bai-Perron test assumes that innovations are independent whereas DCC correlations are de facto smooth, persistent functions of past residuals. As a result, it may overstate test significance and bias break locations, favouring temporary high volatility periods. A stricter LWZ penalty over the standard BIC mitigates this risk, but the algorithm still reaches the maximum allowed number of breaks for all three pairs. It should therefore be interpreted as a descriptive indicator of market shifts rather than statistical evidence of one new prolonged regime.

What the modeling framework cannot tell us

GARCH, EGARCH, DCC, Bai-Perron, are fundamentally descriptive. While they can identify persistent shifts in volatility and correlations, they are unable to establish evidence of structural, underlying economic drivers. The Bond-Equity correlation flip due to inflation spikes or Gold's leverage effect reversal as evidence of its hedging properties, are reasonable hypotheses consistent with the literature, yet, they are not identifiable and quantifiable by the models themselves. A GARCH-MIDAS model would help confirm such hypotheses by integrating various macro variables such as inflation or specific shocks, allowing us to isolate different effects. In conclusion, our analysis provides evidence of the existence of a post-COVID regime shift, but fails to answer questions regarding its duration and underlying economically interpretable drivers.

7 Conclusion

A noticeable shift in the risk structure of US financial markets has occurred since COVID-19. March 2020 saw a spike in overall conditional volatility but mean-reverted quickly. At the univariate level, the evidence is asset-specific: the S&P 500 shows higher persistence and lower short-run reactivity while US IG bonds show higher shock sensitivity and a higher long-run volatility level but not necessarily higher persistence. Gold's GARCH parameters however remain statistically stable. The strongest result lies in the cross-asset interactions. The equity-bond correlation turned positive and the Bai-Perron test identifies March 2020 as a common break month across the three correlation pairs. The shift is driven mainly by correlation rather than relative volatility, as our beta decomposition suggests. The Sharpe ratio of a representative 60/40 portfolio fell by almost 50% and maximum drawdown more than doubled.

We view COVID-19 as a tipping point in the structure of financial risk. It is important to note, however, that we do not regard it as the direct cause of this structural change. The most notable change—the switch in correlation between the equity and bond markets—may be partly due to inflationary pressures and unexpected sharp shifts in interest rates, which could eliminate the ‘flight to safety’ effect that has characterised three decades of negative correlations between the two asset classes. Our dataset does not contain macroeconomic variables such as inflation or key rates so this argument remains subject to interpretation rather than formally tested. Results of the daily rebalanced 60/40 portfolio highlights the fact that the diversification properties of multi-asset-type portfolios are sensitive to structural shifts and that treating historical correlations as a good predictor of future relationships without including the effects of macroeconomic variables can have severe impacts.

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Appendix

The appendix collects the full numerical output that underlies the results presented in the main text: pre-estimation diagnostics, univariate parameter estimates with Bollerslev–Wooldridge robust standard errors, model-selection criteria, residual-adequacy diagnostics, the detailed Bai–Perron break output, the tail-risk panel and a small set of supporting figures. Significance stars throughout the appendix follow the convention *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors are reported in parentheses below each point estimate.

Pre-Estimation Diagnostics

Table 10: Augmented Dickey–Fuller unit-root tests on daily log returns (full sample, H_0 : unit root).

Series	ADF statistic	p -value	Lags used	Decision
S&P 500	−13.862	< 0.001	26	Reject H_0 (stationary)
Gold	−47.353	< 0.001	1	Reject H_0 (stationary)
US IG Bonds	−27.715	< 0.001	4	Reject H_0 (stationary)

Table 11: Ljung–Box test on squared returns (H_0 : no autocorrelation in r_t^2). Rejection at lag 5–20 confirms ARCH effects and motivates GARCH-type modelling.

Asset	Lag 1	Lag 5	Lag 10	Lag 20
S&P 500	757.67 *** ($p < 0.001$)	2,792.41 *** ($p < 0.001$)	4,159.51 *** ($p < 0.001$)	4,833.64 *** ($p < 0.001$)
Gold	160.02 *** ($p < 0.001$)	496.74 *** ($p < 0.001$)	626.13 *** ($p < 0.001$)	736.18 *** ($p < 0.001$)
US IG Bonds	726.31 *** ($p < 0.001$)	1,494.01 *** ($p < 0.001$)	2,087.64 *** ($p < 0.001$)	2,326.60 *** ($p < 0.001$)

Univariate Parameter Estimates with Robust Standard Errors

Tables 12 to 14 report quasi-maximum-likelihood estimates of the GARCH(1,1) and EGARCH(1,1) parameters for each asset across the three samples. Standard errors in parentheses are Bollerslev–Wooldridge sandwich estimates, computed via the `arch` package’s `cov_type='robust'` option. ν denotes the Student- t degrees-of-freedom parameter of the conditional innovation density.

Table 12: S&P 500 – GARCH(1,1) and EGARCH(1,1) parameter estimates with BW-robust standard errors.

	GARCH(1,1)			EGARCH(1,1)		
	Full	Pre-COVID	Post-COVID	Full	Pre-COVID	Post-COVID
ω	0.0252*** (0.0052)	0.0263*** (0.0063)	0.0320*** (0.0096)	−0.0091** (0.0046)	−0.0239*** (0.0077)	−0.0083* (0.0050)
α	0.1635*** (0.0176)	0.1772*** (0.0247)	0.1248*** (0.0222)	0.1715*** (0.0198)	0.1711*** (0.0217)	0.0590* (0.0317)
γ	—	—	—	−0.1967*** (0.0145)	−0.2451*** (0.0260)	−0.1647*** (0.0175)
β	0.8266*** (0.0163)	0.8098*** (0.0224)	0.8523*** (0.0219)	0.9660*** (0.0058)	0.9500*** (0.0094)	0.9827*** (0.0047)
ν	5.33*** (0.44)	4.90*** (0.48)	6.46*** (1.01)	5.83*** (0.54)	5.64*** (0.63)	6.55*** (1.09)
Obs.	4,101	2,548	1,553	4,101	2,548	1,553
Log-Lik.	−5159.1	−2954.8	−2191.5	−5064.9	−2885.8	−2156.2
AIC	10,328.2	5,919.6	4,392.9	10,141.9	5,783.7	4,324.4
BIC	10,359.8	5,948.8	4,419.7	10,179.8	5,818.8	4,356.5
LR vol (%)	25.30	22.53	18.76	13.88	12.50	12.49

Table 13: Gold – GARCH(1,1) and EGARCH(1,1) parameter estimates with BW-robust standard errors.

	GARCH(1,1)			EGARCH(1,1)		
	Full	Pre-COVID	Post-COVID	Full	Pre-COVID	Post-COVID
ω	0.0151*** (0.0051)	0.0076*** (0.0029)	0.0484** (0.0195)	0.0073*** (0.0027)	0.0045* (0.0027)	0.0107* (0.0058)
α	0.0499*** (0.0091)	0.0342*** (0.0068)	0.0802*** (0.0239)	0.1194*** (0.0158)	0.0959*** (0.0146)	0.1479*** (0.0434)
γ	—	—	—	0.0328*** (0.0086)	0.0170* (0.0097)	0.0508*** (0.0159)
β	0.9368*** (0.0120)	0.9575*** (0.0080)	0.8785*** (0.0346)	0.9847*** (0.0039)	0.9903*** (0.0035)	0.9729*** (0.0109)
ν	4.74*** (0.35)	4.75*** (0.44)	5.25*** (0.71)	4.73*** (0.35)	4.71*** (0.44)	5.29*** (0.70)
Obs.	4,236	2,637	1,599	4,236	2,637	1,599
Log-Lik.	−5560.7	−3310.2	−2236.0	−5552.9	−3308.5	−2233.1
AIC	11,131.5	6,630.4	4,482.0	11,117.8	6,628.9	4,478.2
BIC	11,163.2	6,659.8	4,508.9	11,156.0	6,664.2	4,510.4
LR vol (%)	16.91	15.23	17.19	20.18	19.98	19.34

Table 14: US IG Bonds – GARCH(1,1) and EGARCH(1,1) parameter estimates with BW-robust standard errors.

	GARCH(1,1)			EGARCH(1,1)		
	Full	Pre-COVID	Post-COVID	Full	Pre-COVID	Post-COVID
ω	0.0010*** (0.0003)	0.00051** (0.00020)	0.00227*** (0.00086)	−0.0207*** (0.0069)	−0.0197** (0.0083)	−0.0225*** (0.0071)
α	0.0519*** (0.0077)	0.0283*** (0.0053)	0.0705*** (0.0124)	0.0950*** (0.0159)	0.0661*** (0.0114)	0.0736*** (0.0194)
γ	–	–	–	−0.0221*** (0.0073)	−0.0003 (0.0076)	−0.0504*** (0.0096)
β	0.9375*** (0.0095)	0.9639*** (0.0067)	0.9131*** (0.0145)	0.9905*** (0.0028)	0.9922*** (0.0030)	0.9892*** (0.0034)
ν	8.76*** (1.11)	9.97*** (1.93)	8.66*** (1.67)	8.43*** (1.04)	9.55*** (1.76)	10.44*** (2.39)
Obs.	4,267	2,649	1,618	4,267	2,649	1,618
Log-Lik.	−667.0	−53.1	−600.3	−670.4	−58.7	−590.0
AIC	1,344.0	116.2	1,210.6	1,352.8	129.5	1,192.0
BIC	1,375.8	145.6	1,237.5	1,390.9	164.7	1,224.3
LR vol (%)	4.89	4.04	5.91	5.32	4.53	5.59

Model-Selection Criteria (AIC and BIC)

Table 15: Information-criterion comparison of GARCH(1,1) and EGARCH(1,1) by sub-sample. Lower values indicate better fit. The columns Δ AIC and Δ BIC show EGARCH minus GARCH (negative favours EGARCH).

Asset	Sample	GARCH(1,1)		EGARCH(1,1)		EGARCH – GARCH	
		AIC	BIC	AIC	BIC	Δ AIC	Δ BIC
S&P 500	Full	10,328.2	10,359.8	10,141.9	10,179.8	−186.4	−180.1
	Pre-COVID	5,919.6	5,948.8	5,783.7	5,818.8	−135.9	−130.1
	Post-COVID	4,392.9	4,419.7	4,324.4	4,356.5	−68.5	−63.1
Gold	Full	11,131.5	11,163.2	11,117.8	11,156.0	−13.6	−7.3
	Pre-COVID	6,630.4	6,659.8	6,628.9	6,664.2	−1.5	+4.3
	Post-COVID	4,482.0	4,508.9	4,478.2	4,510.4	−3.8	+1.6
US IG Bonds	Full	1,344.0	1,375.8	1,352.8	1,390.9	+8.8	+15.1
	Pre-COVID	116.2	145.6	129.5	164.7	+13.3	+19.2
	Post-COVID	1,210.6	1,237.5	1,192.0	1,224.3	−18.6	−13.2

The two criteria agree on EGARCH for the S&P 500 across every sample and on GARCH for US IG Bonds on the full and pre-COVID samples. For Gold the criteria marginally favour EGARCH; we override that preference on residual-adequacy grounds in Table 17 below.

Residual-Adequacy Diagnostics

The two adequacy diagnostics evaluate the marginal fits along two orthogonal dimensions: distributional adequacy of the innovation density (Kolmogorov–Smirnov) and absence of residual ARCH effects in the variance equation (Ljung–Box on $\hat{z}_t^2$).

Table 16: Kolmogorov–Smirnov test of standardised residuals against the fitted Student- $t(\hat{\nu})$ density. H_0 : standardised residuals follow the fitted distribution; rejection indicates inadequate innovation density.

Asset	Sample	GARCH(1,1)			EGARCH(1,1)		
		$\hat{\nu}$	KS stat	p -value	$\hat{\nu}$	KS stat	p -value
S&P 500	Full	5.33	0.0238	0.019**	5.83	0.0194	0.090*
	Pre-COVID	4.90	0.0272	0.045**	5.64	0.0239	0.108
	Post-COVID	6.46	0.0236	0.347	6.55	0.0251	0.277
Gold	Full	4.74	0.0122	0.553	4.73	0.0104	0.744
	Pre-COVID	4.75	0.0101	0.950	4.71	0.0104	0.936
	Post-COVID	5.25	0.0189	0.609	5.29	0.0153	0.840
US IG Bonds	Full	8.76	0.0242	0.013**	8.43	0.0265	0.005***
	Pre-COVID	9.97	0.0308	0.013**	9.55	0.0306	0.014**
	Post-COVID	8.66	0.0222	0.394	10.44	0.0283	0.147

Table 17: Ljung–Box test on squared standardised residuals $\hat{z}_t^2$. H_0 : no remaining ARCH effects; rejection indicates that the variance equation has not absorbed the conditional heteroskedasticity.

Asset	Sample	Lag	GARCH(1,1)		EGARCH(1,1)	
			Q -stat	p -value	Q -stat	p -value
S&P 500	Full	5	4.06	0.540	7.54	0.183
		10	8.86	0.546	8.12	0.618
		20	15.73	0.734	12.56	0.895
	Pre-COVID	5	3.28	0.657	1.82	0.873
		10	7.28	0.699	7.29	0.698
		20	18.52	0.553	15.84	0.727
	Post-COVID	5	4.11	0.533	7.67	0.176
		10	8.78	0.553	8.95	0.537
		20	13.74	0.843	12.69	0.890
Gold	Full	5	21.66	0.001***	64.30	< 0.001***
		10	23.17	0.010**	67.47	< 0.001***
		20	29.05	0.087*	74.70	< 0.001***
	Pre-COVID	5	43.84	< 0.001***	74.03	< 0.001***
		10	46.12	< 0.001***	77.32	< 0.001***
		20	51.94	< 0.001***	82.71	< 0.001***
	Post-COVID	5	9.74	0.083*	13.05	0.023**
		10	10.49	0.398	13.34	0.205
		20	16.89	0.660	21.98	0.342
US IG Bonds	Full	5	14.45	0.013**	93.43	< 0.001***
		10	27.22	0.002***	147.59	< 0.001***
		20	35.52	0.018**	155.27	< 0.001***
	Pre-COVID	5	5.17	0.395	5.42	0.367
		10	11.05	0.354	13.54	0.195
		20	22.67	0.305	27.47	0.122
	Post-COVID	5	0.64	0.986	3.75	0.585
		10	8.73	0.558	11.97	0.287
		20	12.71	0.890	16.92	0.658

For Gold the EGARCH specification rejects H_0 at every lag on the full and pre-COVID samples ($Q \in [64, 83]$) while GARCH does substantially better ($Q \in [22, 52]$). This is the residual-adequacy criterion that overrides the marginal AIC/BIC preference for EGARCH on Gold; see Section 4.5 of the main text.

Aligned-sample diagnostics for the mixed-marginal DCC input. Tables 18 and 19 report the descriptive moments and Ljung–Box statistics for the standardised residuals of the three selected marginals (EGARCH for S&P 500, GARCH for Gold and US IG Bonds) on the 4,101-observation aligned sample that feeds Step 2 of the DCC estimator.

Table 18: Standardised residuals – descriptive statistics on the aligned sample. Sample mean ≈ 0 and sample variance ≈ 1 confirm that the marginal models adequately capture the volatility dynamics; departures from unit variance would indicate mis-specification of the variance equation.

Statistic	S&P 500	Gold	US IG Bonds
Mean	−0.038	−0.008	−0.022
Std. deviation	1.006	0.992	1.002
Min	−9.479	−6.524	−6.099
25th percentile	−0.561	−0.536	−0.643
Median	0.012	0.013	0.023
75th percentile	0.605	0.566	0.631
Max	3.574	4.535	4.348

Table 19: Ljung–Box test on squared standardised residuals of the mixed-marginal DCC inputs (aligned sample, $T = 4,101$). H_0 : no remaining ARCH effects.

Asset	Marginal	Lag 5		Lag 10		Lag 20	
		Q -stat	p -value	Q -stat	p -value	Q -stat	p -value
S&P 500	EGARCH(1,1)	7.54	0.183	8.12	0.618	12.56	0.895
Gold	GARCH(1,1)	19.15	0.002***	22.06	0.015**	27.57	0.120
US IG Bonds	GARCH(1,1)	13.07	0.023**	24.83	0.006***	30.85	0.057*

Bai–Perron Structural-Break Output

Table 20 gives the full Bai–Perron output that supports the headline break date of $\sim$ March 2020 reported in Section 5.3. The information-criterion column reports the LWZ penalty $IC(m) = T \log(\widehat{\text{SSE}}_m/T) + (m+1)[\log T]^2$; the optimal $\hat{m}$ minimises this column. The minimum segment length is fixed at 126 trading days (≈ 6 months).

Table 20: Bai–Perron mean-shift detection on the three DCC correlation series. $\hat{m}$ is the selected number of breaks with the enforced maximum being 3.

Pair	Segment	Start	End	Mean $\hat{\rho}$
S&P 500 / Gold $\hat{m} = 3$	1	2010-01-05	2014-01-02	+0.154
	2	2014-01-03	2020-03-24	−0.074
	3	2020-03-25	2025-05-05	+0.127
	4	2025-05-06	2026-04-24	−0.010
S&P 500 / Bonds $\hat{m} = 3$	1	2010-01-05	2013-06-12	−0.338
	2	2013-06-13	2014-01-02	+0.018
	3	2014-01-03	2020-03-09	−0.219
	4	2020-03-10	2026-04-24	+0.074
Gold / Bonds $\hat{m} = 3$	1	2010-01-05	2019-05-02	+0.251
	2	2019-05-03	2020-02-28	+0.530
	3	2020-03-02	2024-04-01	+0.343
	4	2024-04-02	2026-04-24	+0.192

Table 21: LWZ information criterion as a function of the number of breaks m . Lower is better; the boxed value identifies the selected $\hat{m}$.

Pair	$m = 0$	$m = 1$	$m = 2$	$m = 3$
S&P 500 / Gold	−13,430.9	−13,780.5	−14,543.0	−14,634.1
S&P 500 / Bonds	−12,416.2	−15,235.4	−15,670.0	−15,957.0
Gold / Bonds	−15,190.4	−16,235.4	−16,759.1	−16,921.9

Tail Risk

Table 22: Historical tail risk by asset and sub-sample. VaR and CVaR are expressed as positive daily-loss percentages; the maximum drawdown is the worst peak-to-trough cumulative loss observed within each sub-sample.

Asset	Period	VaR 1%	CVaR 1%	VaR 5%	CVaR 5%	Max DD (%)
S&P 500	Pre-COVID	2.85	3.63	1.55	2.33	−20.70
	Post-COVID	3.61	5.72	1.84	3.19	−35.86
Gold	Pre-COVID	2.55	3.66	1.49	2.27	−48.27
	Post-COVID	3.42	4.49	1.67	2.66	−23.31
US IG Bonds	Pre-COVID	0.67	0.86	0.40	0.59	−6.51
	Post-COVID	1.05	1.58	0.63	0.94	−21.93

Supporting Figures

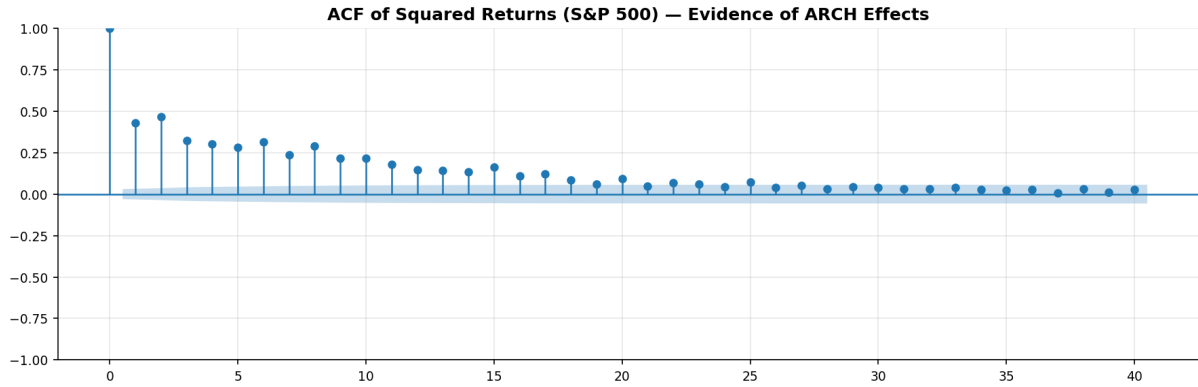


Figure 6: Sample autocorrelation function of squared daily log returns – S&P 500. The slowly-decaying autocorrelation at all displayed lags is the canonical signature of volatility clustering and motivates GARCH-type modelling (see also Table 11).

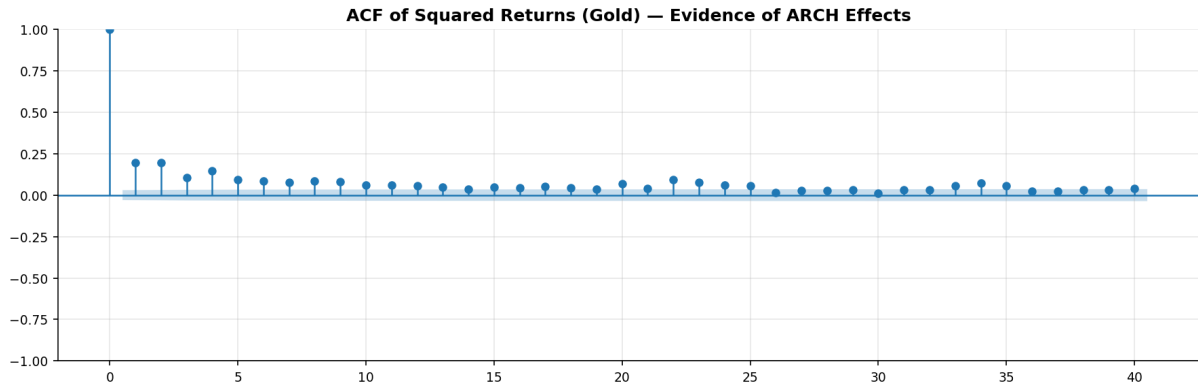


Figure 7: Sample autocorrelation function of squared daily log returns – Gold.

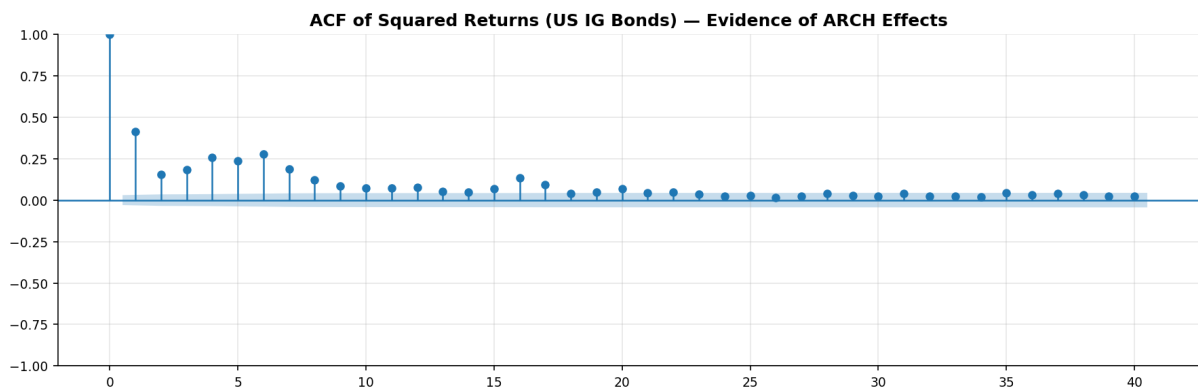


Figure 8: Sample autocorrelation function of squared daily log returns – US IG Bonds.

Gold — ACF of Squared Standardised Residuals ($\hat{z}_t^2$), GARCH(1,1)

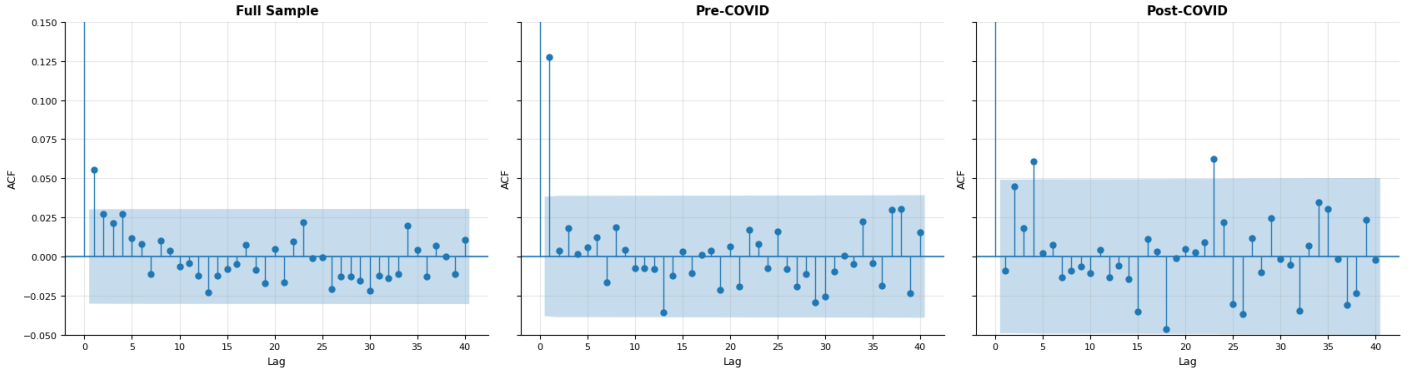


Figure 9: ACF of squared standardised residuals for the Gold GARCH(1,1) model. The remaining autocorrelation is concentrated at short lags, suggesting that the residual ARCH issue is local rather than a persistent long-memory pattern.

S&P 500 — Return Distributions & Standardised Residuals

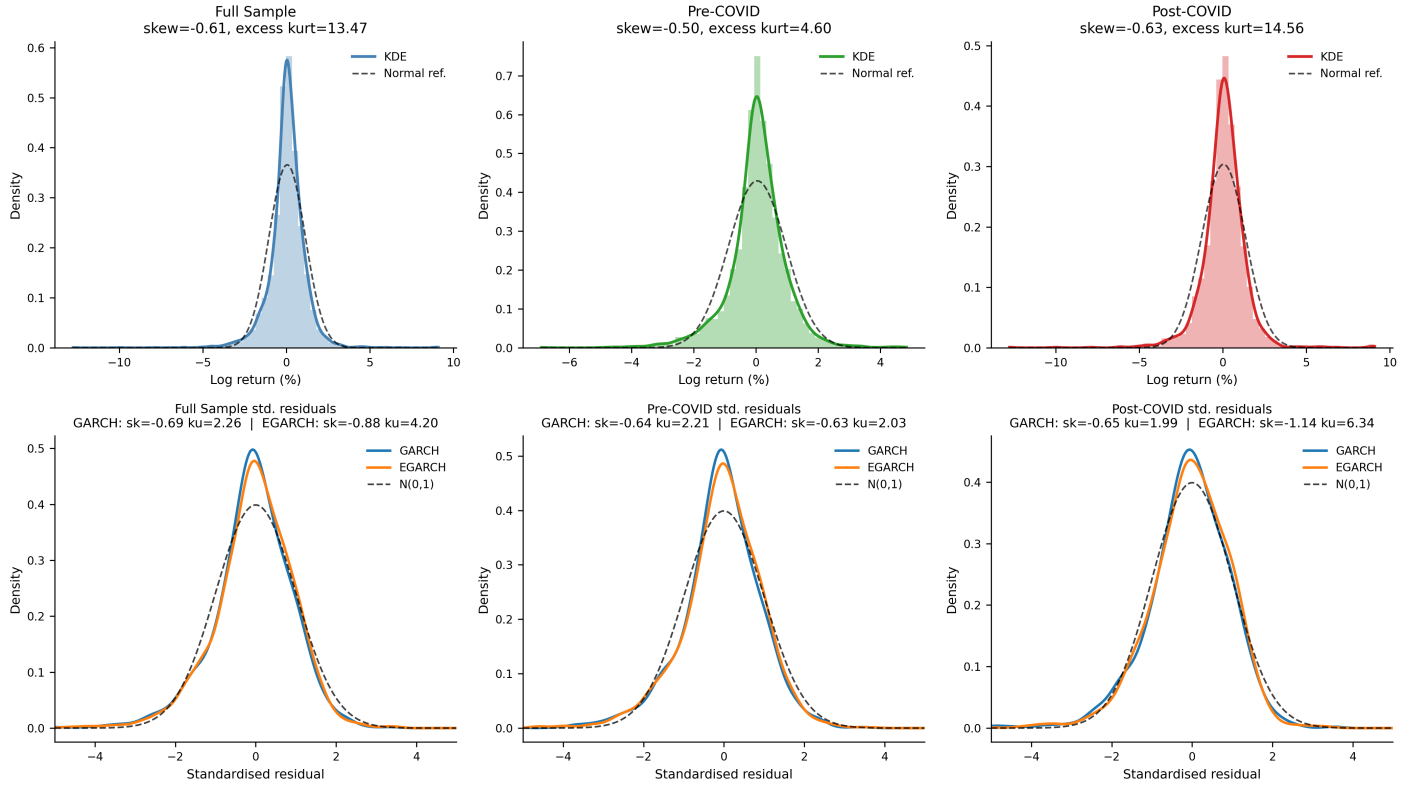


Figure 10: Empirical density of standardised residuals $\hat{z}_t$ versus the standard Normal and the fitted Student- $t(\hat{\nu})$ – S&P 500. The fatter-than-Normal tails motivate the Student- t choice; per-period KS goodness-of-fit tests appear in Table 16.

Gold — Return Distributions & Standardised Residuals

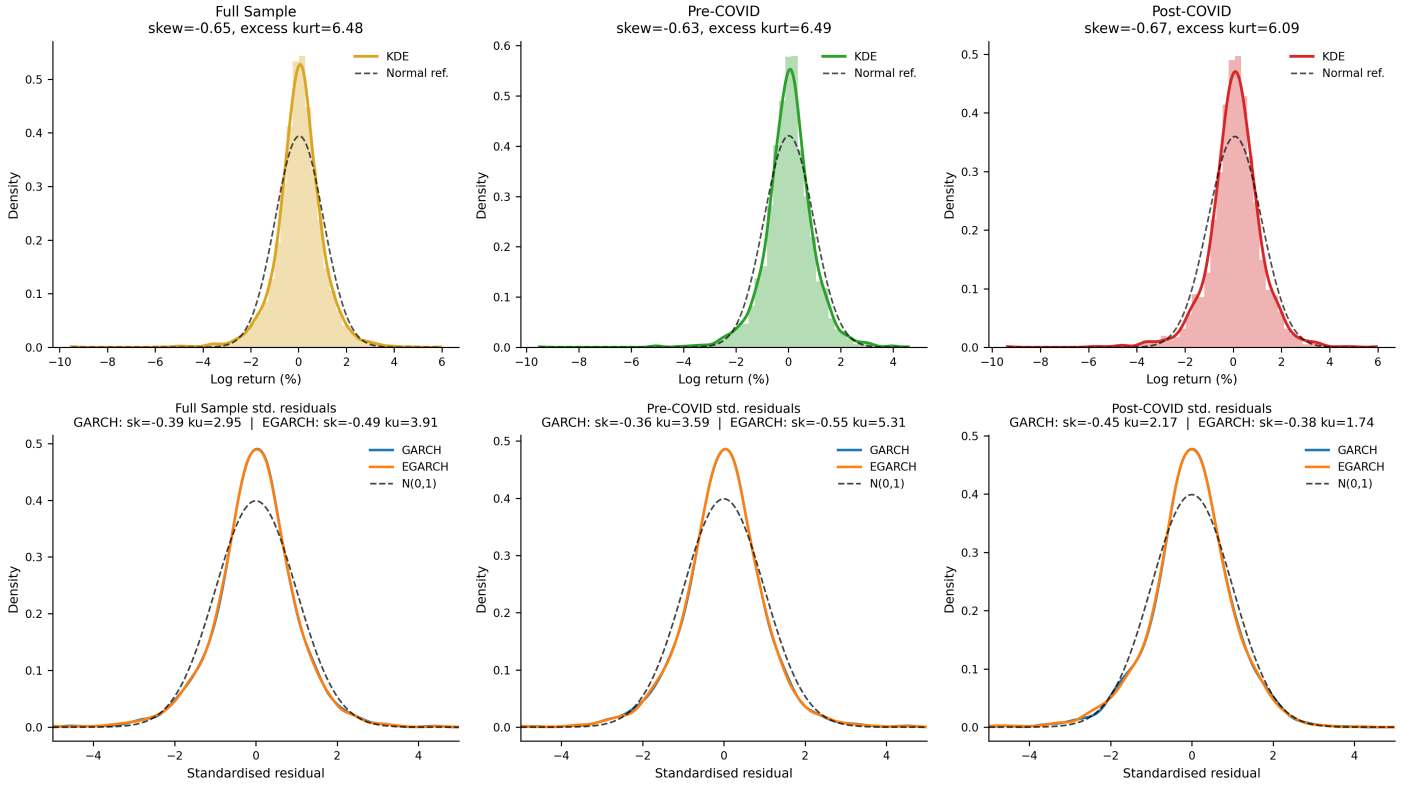


Figure 11: Empirical density of standardised residuals $\hat{z}_t$ versus the standard Normal and the fitted Student- $t(\hat{\nu})$ – Gold.

US IG Bonds — Return Distributions & Standardised Residuals

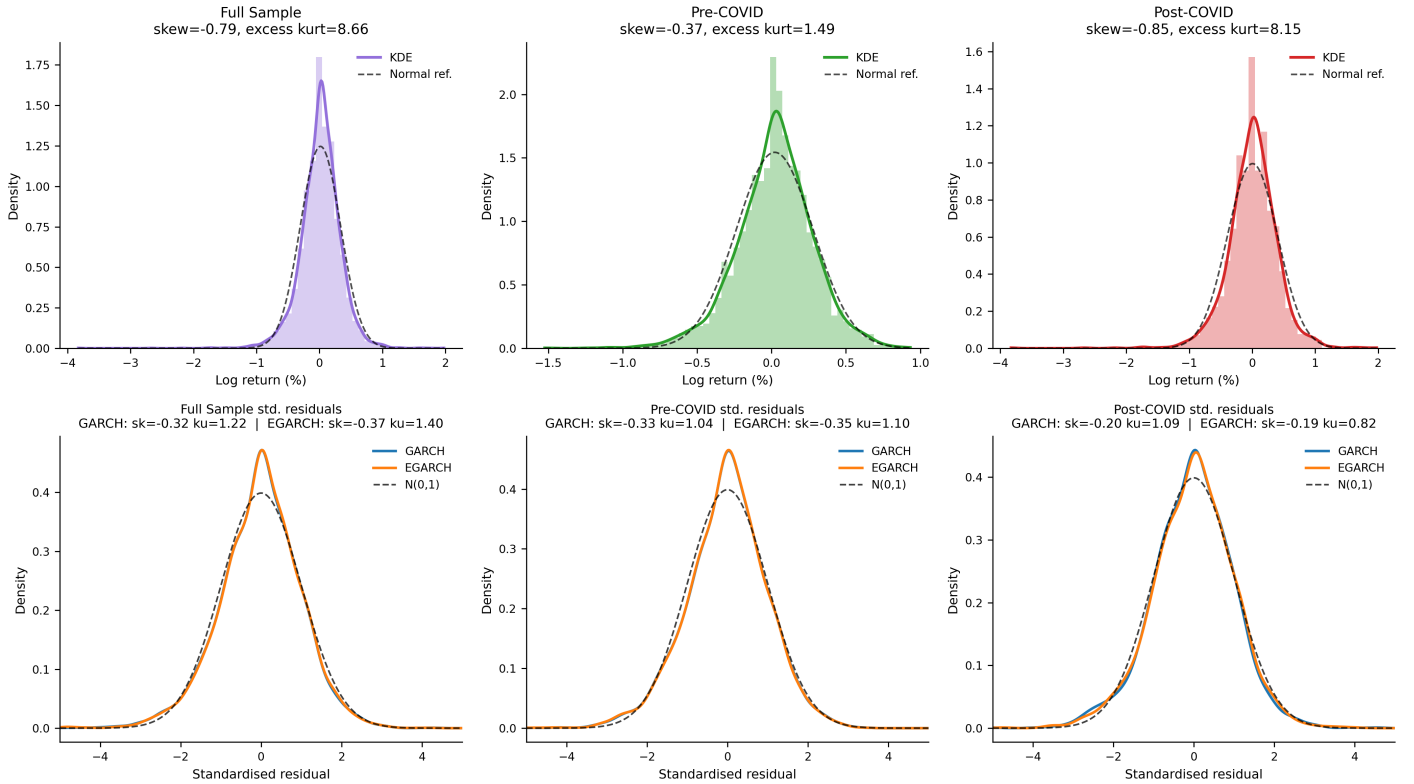


Figure 12: Empirical density of standardised residuals $\hat{z}_t$ versus the standard Normal and the fitted Student- $t(\hat{\nu})$ – US IG Bonds.